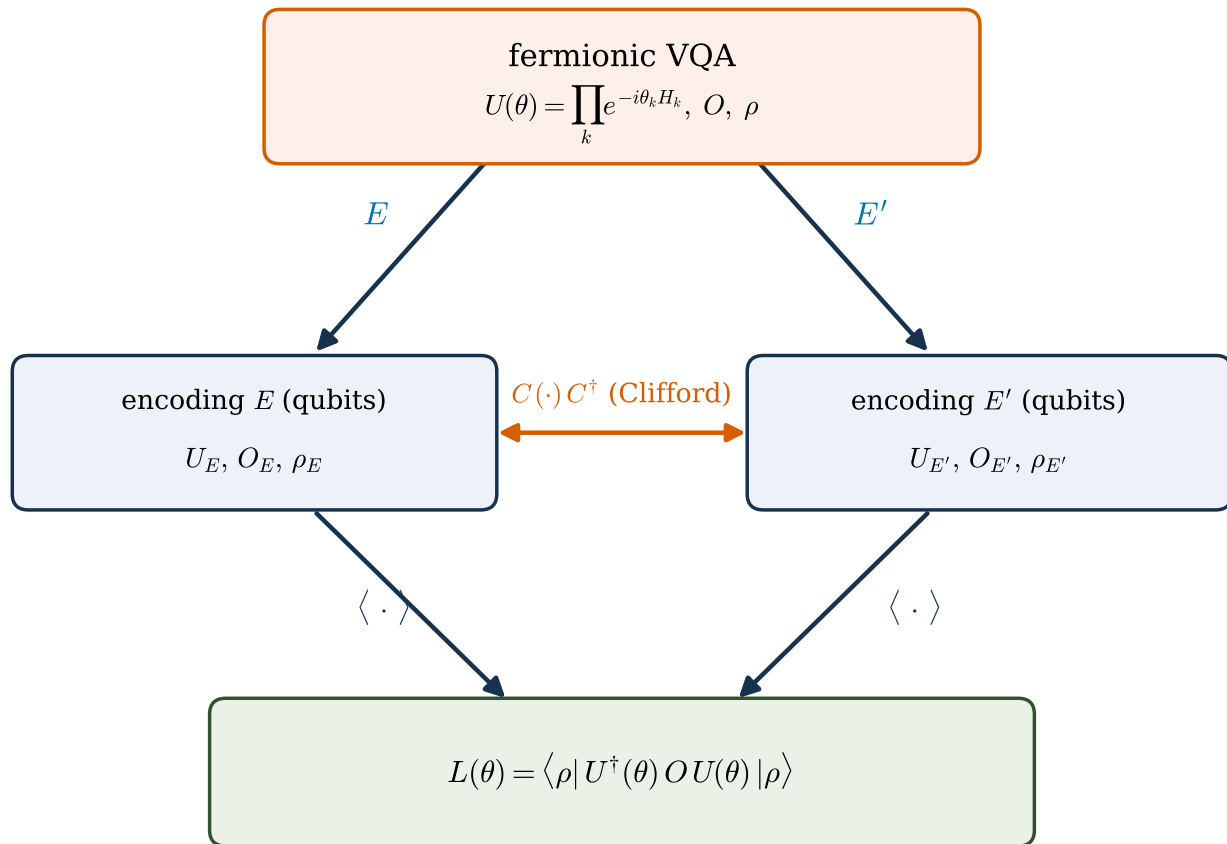


the cost is the same along either path



$L_E(\theta) = L_{E'}(\theta)$ for every encoding $\Rightarrow$ the gradient and its variance are gauge-invariant (Lemma 2)